

(TR-104) PYTHON WITH AI/ML

Training Day 33 Report

Date: 25 February 2026

Topic Covered: Fine-Tuning Transformers, Training Mechanics & Optimization

The thirty-third day of training focused on understanding transformer fine-tuning workflows, dataset processing, Hugging Face Trainer API, model optimization techniques, and evaluation of transformer-based classification models.

- Understood the concept of fine-tuning and how pretrained transformer models are adapted for downstream NLP tasks.
- Studied the difference between pretraining and fine-tuning in transformer architectures.
- Learned how transformer-based classification models internally use contextual embeddings and classification heads for prediction.
- Explored the architecture of `AutoModelForSequenceClassification` and understood the role of logits in classification tasks.
- Loaded and analyzed datasets using the Hugging Face datasets library.
- Worked with dataset splits such as training and testing datasets for supervised learning tasks.
- Created tokenization functions for preprocessing text data before transformer training.
- Applied tokenization mapping efficiently on datasets using batched processing.
- Performed data formatting for PyTorch tensor-based training workflows.
- Initialized pretrained transformer models for sequence classification tasks with appropriate output labels.
- Studied important training parameters including learning rate, batch size, epochs, and weight decay.

- Understood how learning rate affects convergence and model stability during fine-tuning.
- Learned the importance of weight decay for reducing overfitting in transformer training.
- Implemented evaluation metric functions using accuracy score and weighted F1-score.
- Initialized and configured the Hugging Face Trainer API for model training and evaluation.
- Trained transformer models using Trainer and analyzed the training workflow.
- Understood how Trainer internally handles forward pass, backpropagation, optimizer steps, scheduler updates, and evaluation loops.
- Evaluated model performance using validation metrics after training.
- Observed training loss and validation performance for identifying overfitting behavior.
- Learned techniques for reducing overfitting such as reducing epochs and tuning hyperparameters.
- Saved trained transformer models and tokenizers for future inference tasks.
- Reloaded saved models and performed inference on new input sentences.
- Compared transformer-based models with classical NLP approaches such as TF-IDF and Logistic Regression.
- Analyzed differences between classical NLP and transformer models in terms of accuracy, contextual understanding, model size, and training complexity.

GitHub Link:

https://github.com/JaspinderKaurWalia26/Transformers/tree/main/fine_tuning_transformer

Blog Link:

<https://medium.com/@jaspinderkaurwalia855/fine-tuning-transformers-from-pretrained-knowledge-to-task-mastery-02b07a2c0252>

Overall, the training helped in strengthening understanding of transformer fine-tuning workflows, Hugging Face Trainer API, dataset processing, training

optimization, evaluation strategies, and practical transformer-based NLP model development.